

TCS349: Responsible and Explainable AI

Unit 2, Lecture 3

Transparency Tools: Model Cards, Datasheets & Audit Logs

The templates and records that turn transparency from a principle into a practice - what to document, for a model, its data, and the system as a whole.

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Model Cards



Datasheets



System Cards



Audit Logs

2. What Is a Model Card?

A model card is a short, standardised document describing an AI model - what it does, how it was built, and where it should and should not be used.

It was introduced to give consumers of a model the information needed to decide whether it fits their use case.

Model cards are usually one to two pages long, written for a mixed technical and non-technical audience.

Publishing a model card is one of the most common ways organisations demonstrate transparency in practice.

3. Model Card - Typical Sections

Section	What It Contains
Model Details	Owner, version, date, model type and intended purpose
Intended Use	Primary use cases and users the model was designed for
Performance	Accuracy and fairness metrics, including subgroup-wise results
Limitations	Known weaknesses, out-of-scope uses and edge cases
Ethical Considerations	Risks identified and mitigations applied

4. Model Card - Worked Mini-Example

Model Details: Loan-approval classifier, version 2.1, trained March 2026.

Intended Use: Screening personal loan applications for a retail bank; not intended for business or mortgage loans.

Performance: Overall accuracy 84%; TPR 80% for Group A versus 50% for Group B - flagged as a known limitation.

Limitations: Lower recall for Group B applicants; not validated on applicants under 21.

5. What Is a Datasheet for Datasets?

A datasheet documents a dataset the same way a model card documents a model - its origin, composition and appropriate uses.

It answers a simple question: can I trust this data enough to build or evaluate a model on it?

Datasheets borrow their name from the idea that electronics components ship with a datasheet - datasets deserve the same discipline.

A dataset without a datasheet is difficult to audit for bias, consent or representativeness later.

6. Datasheet - Typical Sections

Section	What It Contains
Motivation	Why the dataset was created and by whom
Composition	What the data represents, and who or what it covers
Collection Process	How and when the data was gathered, and any consent obtained
Preprocessing	Cleaning, labelling or transformations applied before use
Recommended Uses	Appropriate uses, and any uses the creators advise against

7. What Is a System Card?

A system card documents an entire AI-powered product or service, not just one underlying model.

Real products often combine multiple models, business rules and human review steps - a system card captures the whole pipeline.

It typically covers how the system is monitored, updated and governed after deployment, not only how it was built.

System cards are the natural documentation layer for the trustworthy AI lifecycle introduced in Lecture 10.

8. Model Card vs Datasheet vs System Card

Dimension	Model Card	Datasheet	System Card
Documents	One trained model	One dataset	The full product or service
Typical Author	ML engineer or data scientist	Data collection team	Product or governance team
Key Question	Should I use this model, and how?	Can I trust this data?	How does the system behave, and how is it governed?

9. Documentation Beyond Cards

Design documents: capture the reasoning behind key architectural and data decisions before a system is built.

Risk assessments: formally identify and rate potential harms before deployment, alongside planned mitigations.

Change logs: record what changed between versions of a model or system, and why.

Together with cards and datasheets, these form a full documentation trail across a system's life.

10. What Is an Audit Log?

An audit log is a running record of events in an AI system - inputs, outputs, model version and key decisions - stored as they occur.

It is the mechanism that makes the traceability introduced in Lecture 11 actually verifiable.

Audit logs are typically append-only, meaning past entries cannot be silently edited or deleted.

Without an audit log, a "we can explain this decision" claim usually cannot be checked after the fact.

11. Audit Log - What It Should Capture

Field	Example Value
Timestamp	2026-03-14 10:42:17
Model Version	loan-classifier v2.1
Input Reference	Application ID 48213
Output / Decision	Rejected
Reviewer (if any)	Escalated to human reviewer R. Sharma

12. Putting It Together - Documentation Across the Lifecycle

Design stage: design documents and early risk assessments.

Data stage: datasheets for every dataset used in training or evaluation.

Development and testing: model cards summarising performance and limitations.

Deployment and monitoring: system cards and audit logs that stay current for as long as the system is live.

13. Common Pitfalls in Transparency Documentation

Writing a model card once at launch and never updating it as the model is retrained.

Omitting subgroup-wise performance results, hiding the exact disparities Unit I taught you to detect.

Using audit logs that can be edited after the fact, undermining their value as evidence.

Writing documentation only for engineers, leaving other stakeholders unable to understand it.